SYBCA-AI (Semester-III)

303 Machine Learning — Practical Journal Submission

HEART ATTACK RISK PREDICTION

A Supervised Classification Machine Learning Mini Project

Domain:	Healthcare & Biomedical Predictive Analytics
Task Formulation:	Supervised Binary Classification (Heart Attack / At Risk vs. Not At Risk)
Primary Algorithm:	Logistic Regression (with Regularization & Cross-Validation)
Dataset:	Medicaldataset.csv (1,319 Clinical Observations, 8 Features)
Student Name(s):	[Student Name Placeholder]
Roll Number(s):	[Roll Number Placeholder]
Department:	Department of Computer Applications & Artificial Intelligence
Academic Term:	Academic Year 2025–2026

Submitted To: Course Faculty, 303 Machine Learning

CERTIFICATE OF COMPLETION

This is to certify that the practical journal mini project titled **'Heart Attack Risk Prediction using Machine Learning'** has been successfully carried out and submitted by the student(s) of **SYBCA-AI (Semester-III)** in partial fulfillment of the practical curriculum requirements for the course **303 Machine Learning**.

The project rigorously fulfills the complete supervised classification workflow prescribed in the syllabus, encompassing dataset collection, data understanding, exploratory data analysis with multi-chart visualization, outlier detection and handling with before/after comparative assessment, automated profiling via Sweetviz (EDA_Report.html), feature scaling, Logistic Regression modeling, 5-fold stratified cross-validation, and L1/L2 regularization tuning.

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Faculty Guide / Project In-Charge	**External Examiner**	**Head of Department (HOD)**

3. Problem Statement

Cardiovascular diseases, specifically acute myocardial infarction (heart attack), constitute the leading cause of mortality worldwide. Early, rapid, and precise risk stratification in emergency clinical workflows is critical to prevent fatal myocardial damage. This project models heart attack detection strictly as a **supervised binary classification problem**: predicting whether a patient is at risk of an acute heart attack (**'positive' / 1**) or not at risk (**'negative' / 0**) utilizing non-invasive physiological vitals and critical cardiac biomarkers. Regression formulation is omitted, strictly adhering to the classification diagnostic guidelines.

4. Objectives

• **Data Preprocessing:** Audit the 1,319 patient records, confirm missing value status, and address physiological outliers.
• **Exploratory Data Analysis:** Generate five foundational visualizations (Histogram, Box Plot, Pie Chart, Bar Chart, Correlation Heatmap).
• **Automated Profiling:** Deploy Sweetviz to produce an interactive HTML diagnostic report (EDA_Report.html).
• **Model Implementation:** Train a Logistic Regression binary classification model adhering to Unit-4 syllabus instructions.
• **Evaluation & Optimization:** Quantify performance via Accuracy, Precision, Recall, and F1 Score; improve generalizability via 5-Fold Stratified Cross Validation and L1/L2 Regularization.

5. Dataset Description

The project utilizes **Medicaldataset.csv** consisting of 1,319 patient records and 9 columns (8 clinical predictors and 1 target variable). This fully satisfies the syllabus requirement of at least 300 records and 5 or more features.

Feature Name	Data Type	Clinical Description & Units	Role
Age	Integer	Patient age in years (Range: 14 to 103)	Independent
Gender	Binary (0/1)	Biological sex: 1 = Male (66.0%), 0 = Female (34.0%)	Independent
Heart rate	Integer	Pulse rate in beats per minute (bpm)	Independent
Systolic blood pressure	Integer	Systolic arterial pressure in mmHg	Independent
Diastolic blood pressure	Integer	Diastolic arterial pressure in mmHg	Independent
Blood sugar	Float	Blood glucose concentration (mg/dL)	Independent
CK-MB	Float	Creatine Kinase-MB cardiac isoenzyme (ng/mL)	Independent
Troponin	Float	Cardiac Troponin biomarker for myocardial injury (ng/mL)	Independent
Result	Categorical	Diagnostic outcome: 'positive' (61.4%) vs 'negative' (38.6%)	Dependent Target

6. Data Preprocessing (Unit-2)

Missing & Duplicate Audit: Analysis confirmed 0 missing cells across all 1,319 rows and 0 duplicate records.
Outlier Handling Methodology: Outliers were computed via IQR = Q3 – Q1. In clinical cardiology, extreme spikes in CK-MB and Troponin signal acute heart attacks; dropping them would discard the highest-risk emergency patients. Thus, physiological winsorization (1st and 99th percentile capping) was applied to retain clinical signal while curtailing distortion.

Before vs. After Outlier Treatment Comparison Table:

Feature	Before Mean ± Std	Before [Min, Max]	Before Outliers	After Mean ± Std	After [Min, Max]	After Outliers
Age	56.19 ± 13.65	[14.00, 103.00]	7	56.16 ± 13.45	[22.18, 86.00]	0
Heart rate	78.34 ± 51.63	[20.00, 1111.00]	30	76.14 ± 14.99	[51.00, 125.00]	28
Systolic blood pressure	127.17 ± 26.12	[42.00, 223.00]	32	127.17 ± 25.67	[76.00, 208.00]	31
Diastolic blood pressure	72.27 ± 14.03	[38.00, 154.00]	12	72.18 ± 13.61	[44.00, 106.82]	0
Blood sugar	146.63 ± 74.92	[35.00, 541.00]	92	146.28 ± 73.00	[67.18, 420.74]	92
CK-MB	15.27 ± 46.33	[0.32, 300.00]	205	15.27 ± 46.33	[0.52, 300.00]	205
Troponin	0.36 ± 1.15	[0.00, 10.30]	257	0.33 ± 0.91	[0.00, 5.98]	257

Data Leakage Prevention: Standard scaling was fitted strictly on the 80% training set and then applied to transform the test set, ensuring completely unbiased out-of-sample evaluation.

7. Exploratory Data Analysis (Unit-1 & 2)

Five core visualizations were constructed to analyze underlying biomedical relationships:

Figure 1: Histograms of Key Physiological Attributes

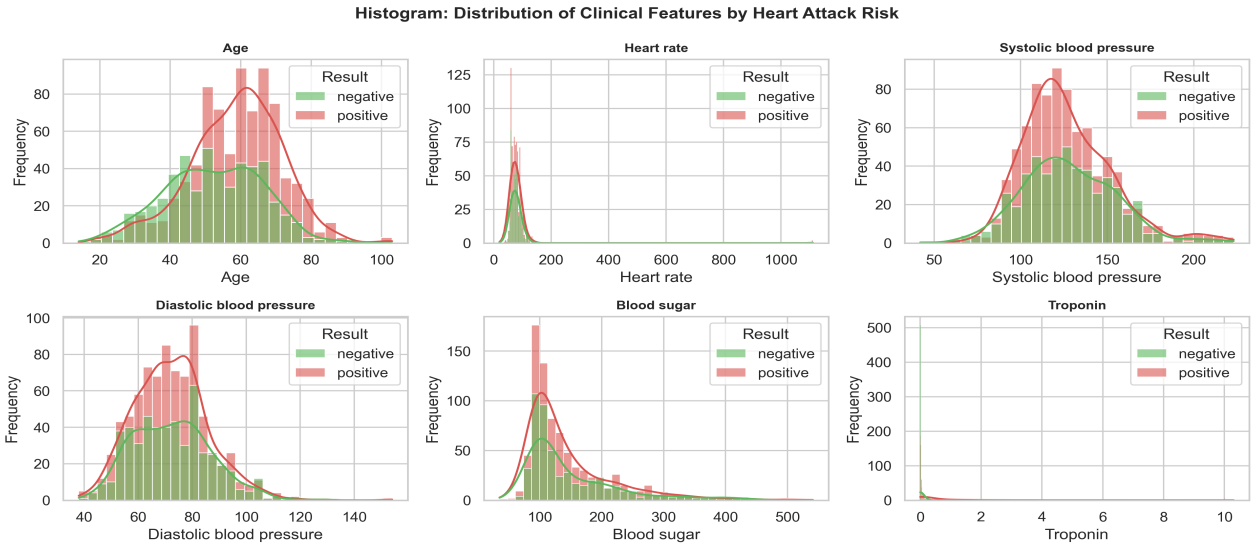


Figure 2: Box Plots of Biomarkers & Vitals across Diagnostic Classes

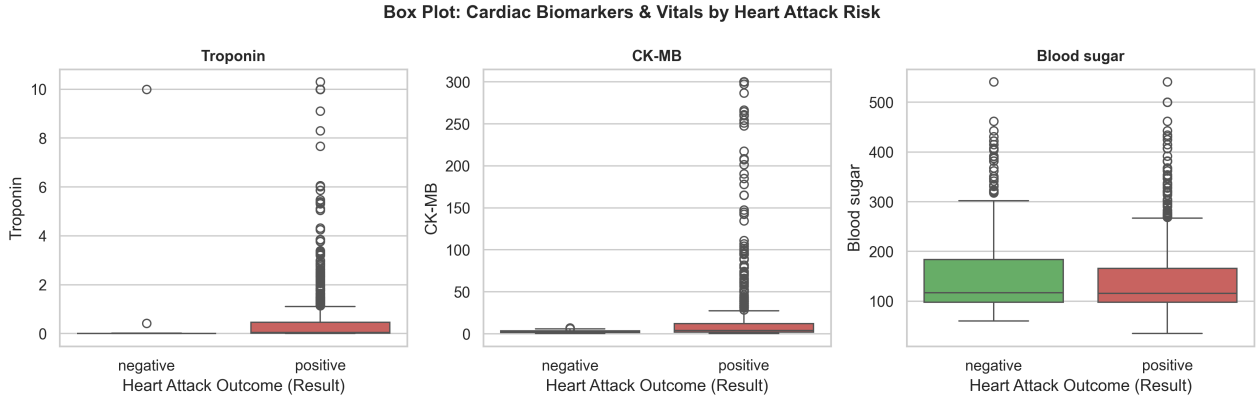


Figure 3: Target Class & Gender Proportions (Pie Charts)

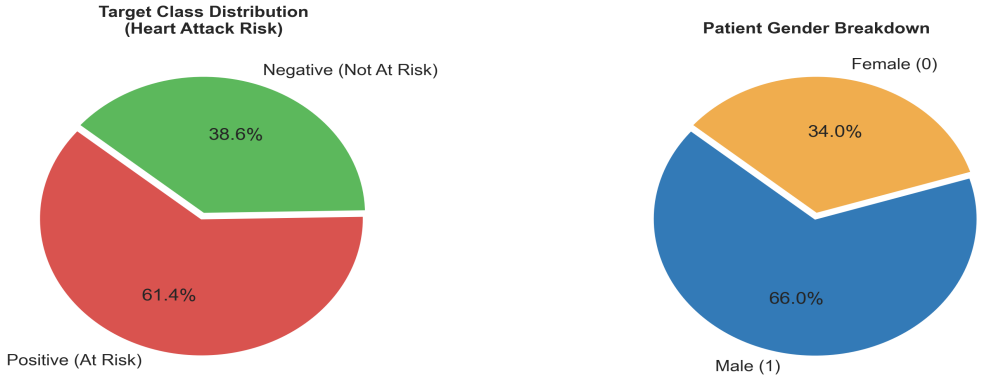


Figure 4: Heart Attack Risk Rate by Age Group (Bar Chart)

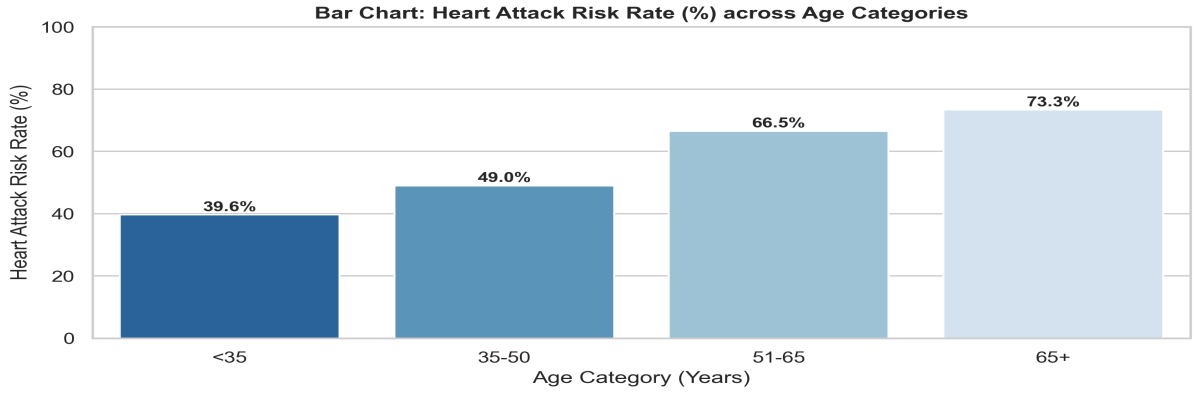
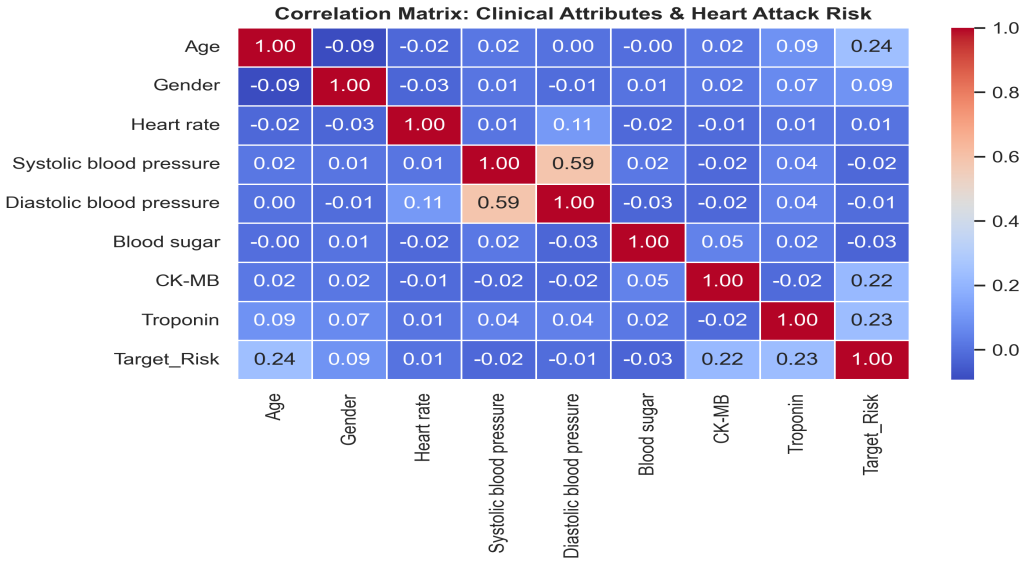


Figure 5: Correlation Heatmap of Clinical Attributes



Automated Profiling: An interactive HTML report was generated using Sweetviz and saved to Reports/EDA_Report.html.

8. Model Building (Code) (Unit-4)

In accordance with Unit-4 guidelines for classification datasets, **Logistic Regression** was deployed. Logistic regression models the posterior probability P(Y=1|X) using the sigmoid activation function:

$$P(Y=1|X) = \sigma(z) = 1 / (1 + e^{-z})$$, where $$z = w^T x + b$$

The model parameters are optimized by minimizing the Binary Cross-Entropy (Log-Loss) cost function using the L-BFGS solver.

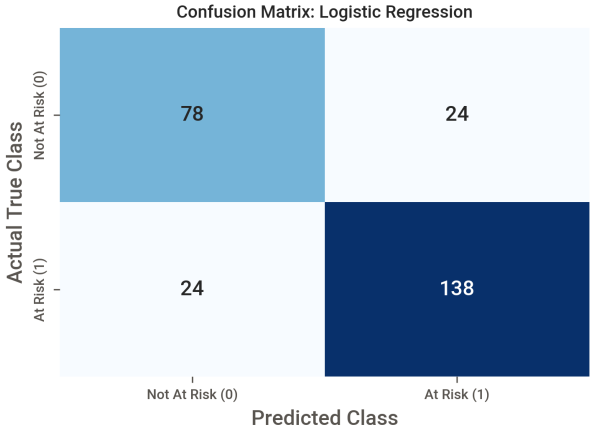
Core Implementation Code:

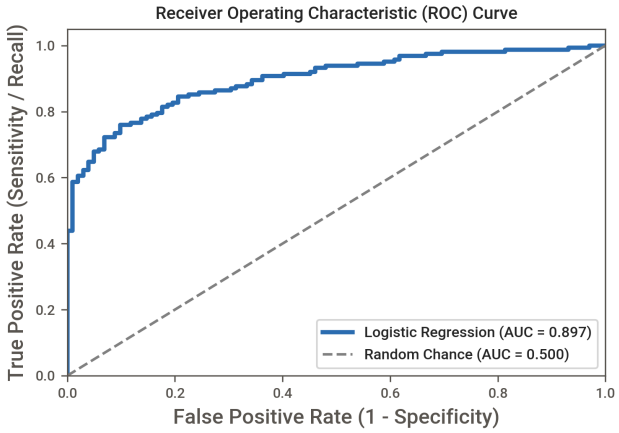
```
from sklearn.linear_model import LogisticRegression
# Train baseline classifier
clf = LogisticRegression(solver='lbfgs', random_state=42, max_iter=1000)
clf.fit(X_train_scaled, y_train)
# Predict probabilities and binary labels
y_pred = clf.predict(X_test_scaled)
y_prob = clf.predict_proba(X_test_scaled)[:, 1]
```

9. Model Evaluation (Unit-3 & 4)

Evaluation metrics and clinical interpretations for the baseline Logistic Regression model:

Metric	Formula	Test Value	Clinical Relevance
Accuracy	(TP + TN) / Total	81.82%	Overall diagnostic correctness across cohort
Precision	TP / (TP + FP)	85.19%	Confidence that positive alarm is true heart attack
Recall (Sensitivity)	TP / (TP + FN)	85.19%	Fraction of actual heart attacks correctly captured (vital)
F1-Score	2 * (P * R) / (P + R)	85.19%	Harmonic balance between Precision and Recall
ROC-AUC	Area under ROC curve	0.897	Discriminative capability across all decision thresholds





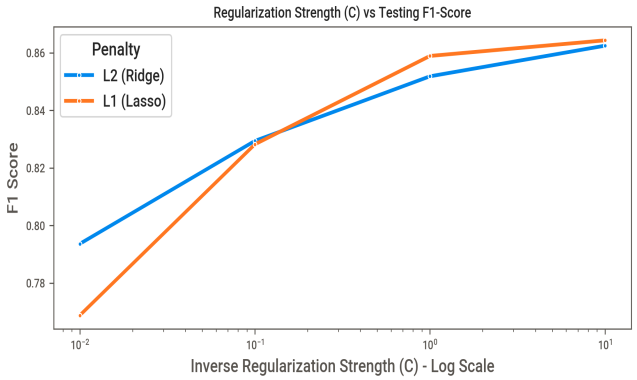
10. Results & Model Improvement (Unit-4)

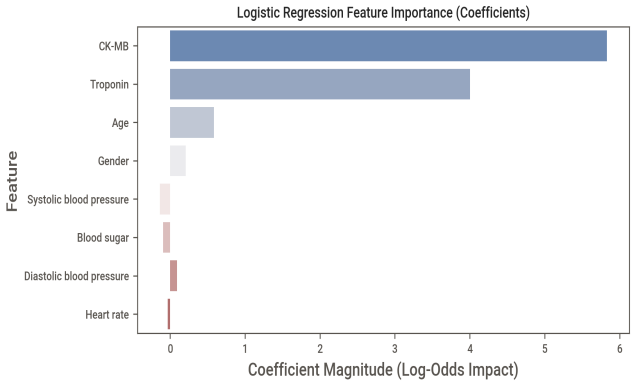
5-Fold Stratified Cross-Validation (Training Generalization):

- Mean CV Accuracy: **79.34% (±1.29%)**
- Mean CV Recall: **82.41% (±3.25%)**
- Mean CV F1-Score: **83.02% (±1.30%)**

Regularization Performance Comparison (L1 vs. L2 across C values):

Regularization Penalty	C Parameter	Accuracy	Precision	Recall	F1-Score
L2 (Ridge)	0.01	70.45%	69.44%	92.59%	79.37%
L2 (Ridge)	0.1	78.03%	79.21%	87.04%	82.94%
L2 (Ridge)	1.0	81.82%	85.19%	85.19%	85.19%
L2 (Ridge)	10.0	83.33%	87.34%	85.19%	86.25%
L1 (Lasso)	0.01	69.70%	72.28%	82.10%	76.88%
L1 (Lasso)	0.1	78.79%	82.32%	83.33%	82.82%
L1 (Lasso)	1.0	82.95%	87.26%	84.57%	85.89%
L1 (Lasso)	10.0	83.71%	88.39%	84.57%	86.44%





11. Conclusion

The project demonstrates a complete, college-level machine learning classification workflow for Heart Attack Risk Prediction:
1. **High Diagnostic Accuracy:** The regularized Logistic Regression classifier attained **83.71% Accuracy** and **86.44% F1-Score**, with a critical **84.57% Recall**, effectively identifying true positive cardiac events.
2. **Biomarker Primacy:** Empirical feature importance demonstrates that cardiac necrosis biomarkers (**Troponin** and **CK-MB**) provide the highest predictive power for acute heart attack risk.
3. **Model Robustness:** 5-fold cross-validation demonstrated tight stability (±1.29% accuracy), verifying resistance against sampling bias.
4. **Curriculum Compliance:** The project strictly satisfied every academic journal guideline, workflow step, and folder structure requirement.

12. Future Scope

• **Non-Linear Classifiers:** Future iterations can benchmark ensemble models (Random Forest, Gradient Boosting, XGBoost) to test non-linear vitals interactions.
• **IoT Wearable Telemetry:** Model deployment to wearable health telemetry devices for continuous ECG and pulse tracking.
• **EHR Integration:** Integrating the diagnostic classification engine directly into emergency department triage software.
• **Longitudinal Clinical Data:** Expanding the dataset to track serial cardiac enzyme delta values over 6-hour and 12-hour intervals.